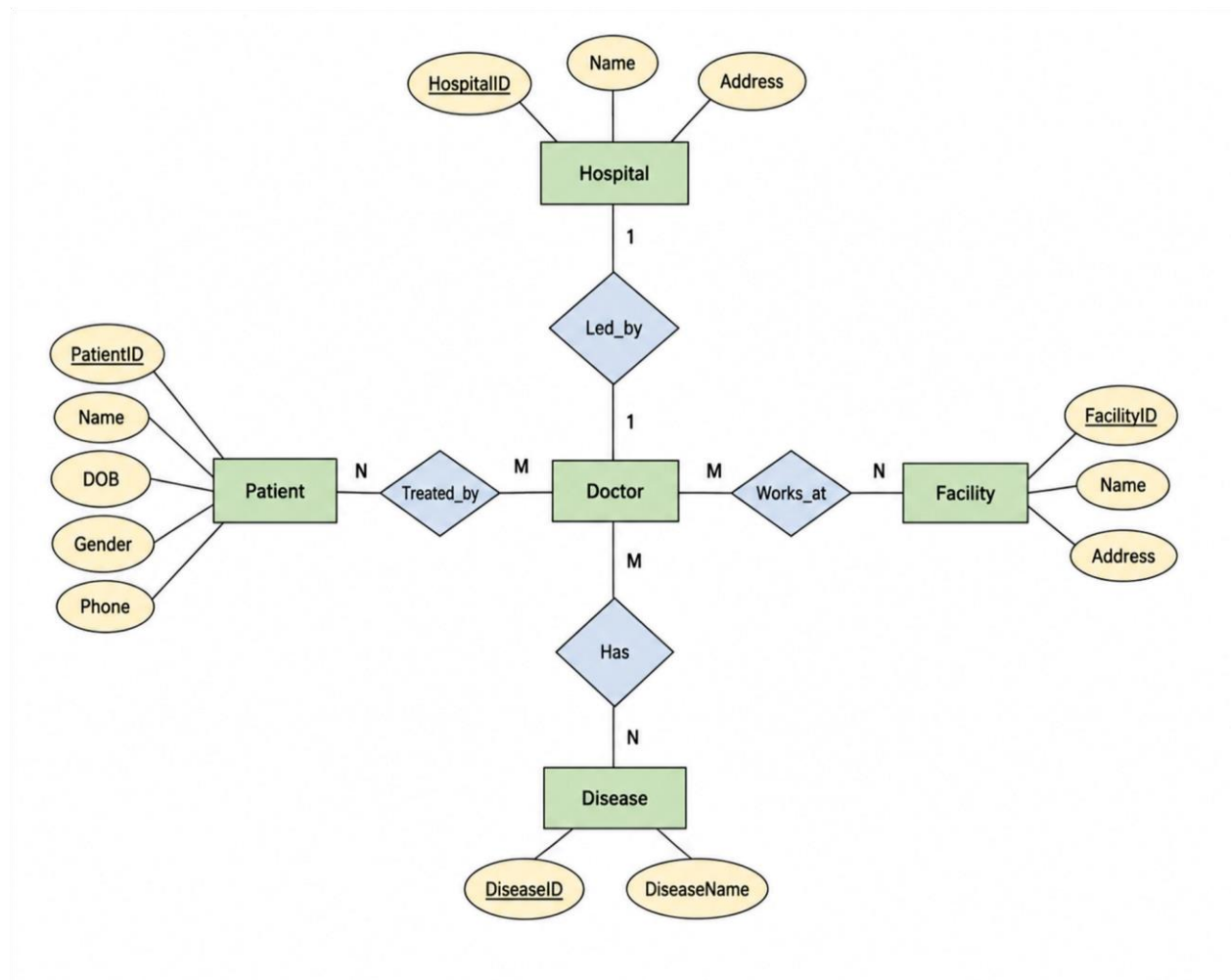


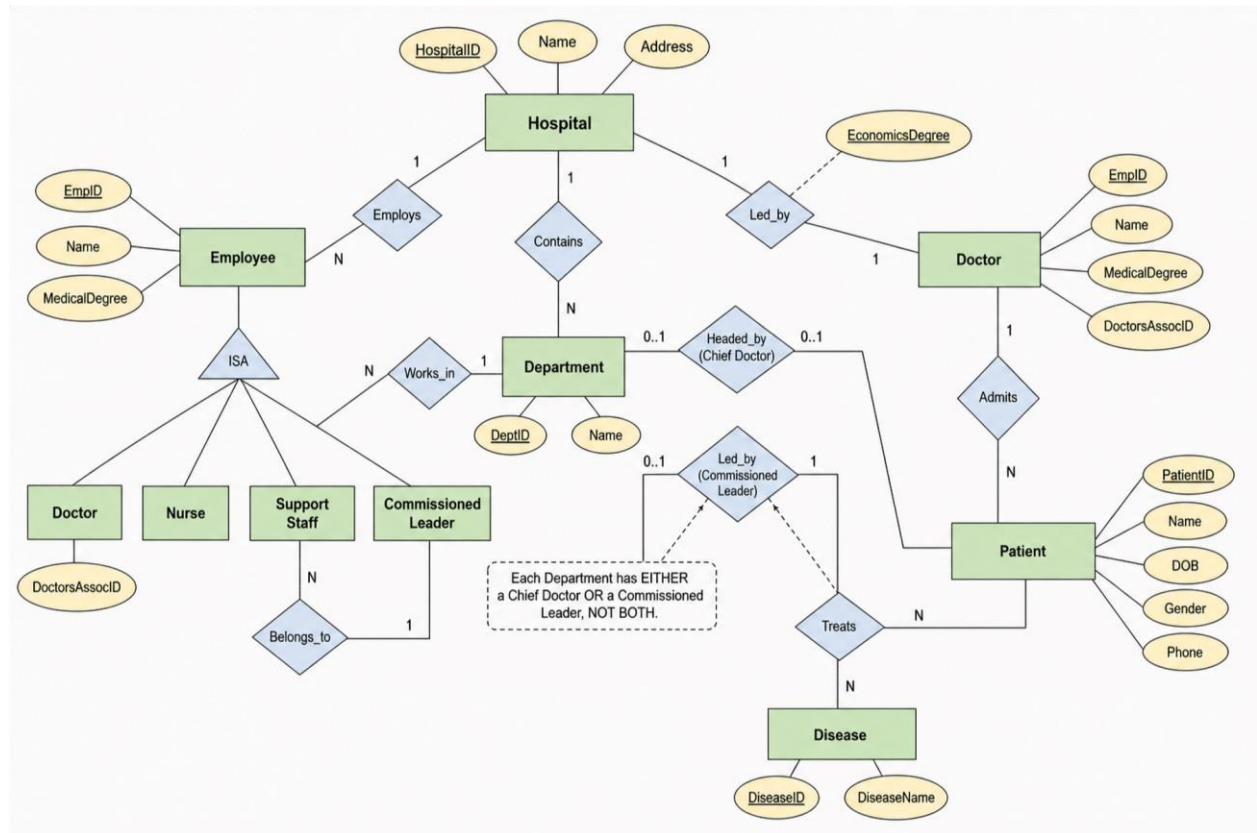
DBMS LAB ASSIGNMENT - 2

Vamshi Anirudh
24MAB0A42

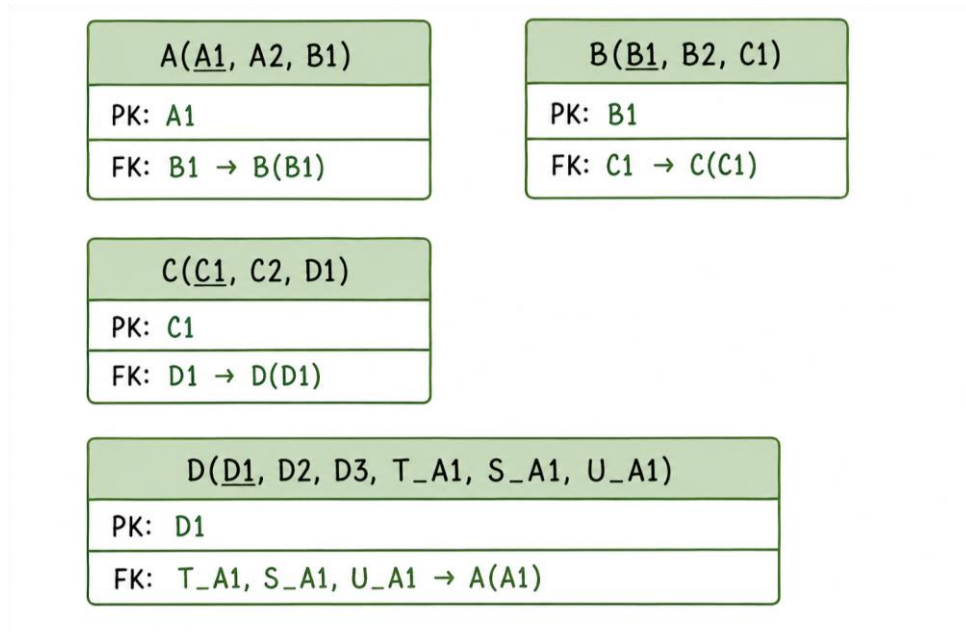
Q1



Q2

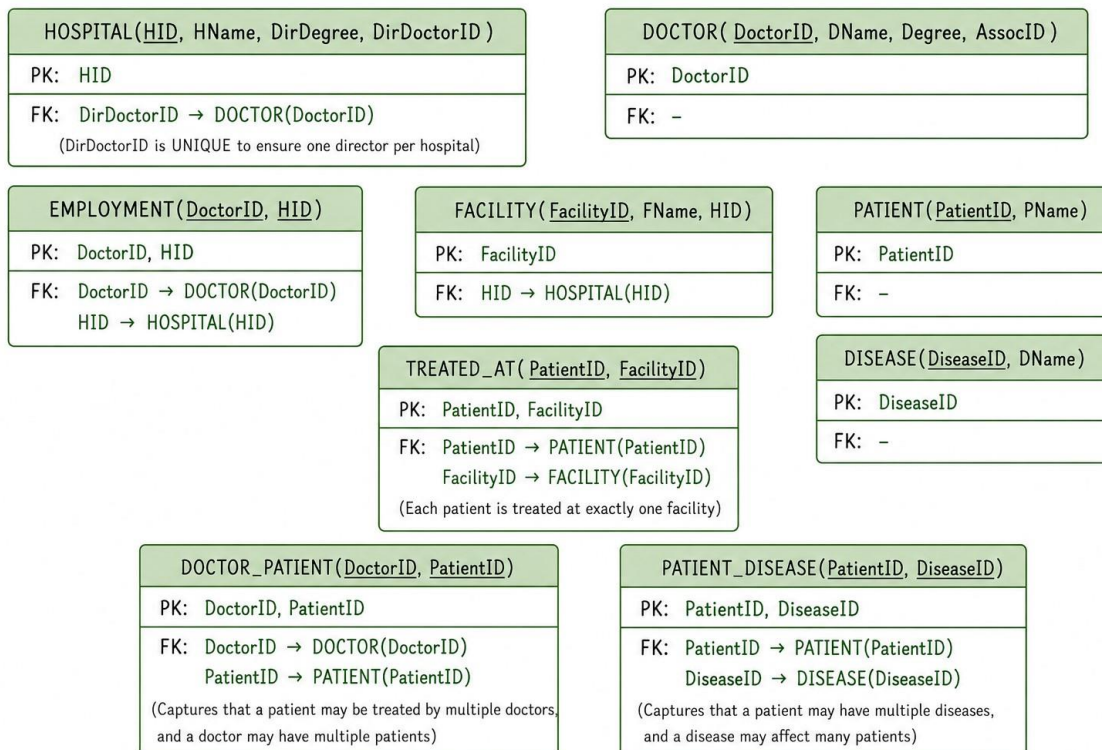


Q3



Q4

1)



2)

HOSPITAL(<u>HID</u>, HName, DirDoctorID) PK: HID FK: DirDoctorID → DOCTOR(DirectorID) (DirDoctorID is UNIQUE to ensure one director per hospital)	DOCTOR(<u>DirectorID</u>, DName, Degree, AssocID) PK: DirectorID FK: AssocID → DOCTORS_ASSOCIATION(AssocID)	DOCTORS_ASSOCIATION(<u>AssocID</u>, AssocName) PK: AssocID FK: -	
HOSPITAL_DOCTOR(<u>DoctorID</u>, <u>HID</u>) PK: DoctorID, HID FK: DoctorID → DOCTOR(DirectorID) HID → HOSPITAL(HID) (A doctor is employed by a maximum of three hospitals - business rule to be enforced)	DEPARTMENT(<u>DeptID</u>, DName, HID, LeaderDoctorID, IsCommissioned) PK: DeptID FK: HID → HOSPITAL(HID) LeaderDoctorID → DOCTOR(DirectorID) IsCommissioned: 'Y' if department leader is commissioned (not a chief doctor), 'N' otherwise (If IsCommissioned - 'N' then LeaderDoctorID must be a chief doctor)	DEPARTMENT_CHIEF(<u>DeptID</u>, <u>DoctorID</u>) PK: DeptID, DoctorID FK: DeptID → DEPARTMENT(DepartmentID) DoctorID → DOCTOR(DirectorID) (Stores all chief doctors of a department. If no record exists for a department, then it has a commissioned leader)	
EMPLOYEE(<u>EmpID</u>, EmpName, Degree) PK: EmpID FK: - (All employees have a medical degree and are not employed by any other hospital)	DOCTOR_EMPLOYEE(<u>EmpID</u>, <u>AssocID</u>) PK: EmpID FK: EmpID → EMPLOYEE(EmpID) AssocID → DOCTORS_ASSOCIATION(AssocID) (One-to-one between a doctor (employee) and their doctors' association ID)	NURSE (<u>EmpID</u>, <u>DeptID</u>) PK: EmpID FK: EmpID → EMPLOYEE(EmpID) DeptID → DEPARTMENT(DepartmentID)	SUPPORT_STAFF(<u>EmpID</u>, <u>HID</u>) PK: EmpID FK: EmpID → EMPLOYEE(EmpID) HID → HOSPITAL(HID) (Support staff belongs directly to the hospital)
PATIENT(<u>PatientID</u>, PName, <u>HID</u>) PK: PatientID FK: HID → HOSPITAL(HID)	PATIENT_DEPARTMENT(<u>PatientID</u>, <u>DeptID</u>) PK: PatientID, DeptID FK: PatientID → PATIENT(PatientID) DeptID → DEPARTMENT(DepartmentID) (A patient may be treated in multiple departments until cured)	PATIENT_DISEASE(<u>PatientID</u>, <u>DiseaseID</u>) PK: PatientID, DiseaseID FK: PatientID → PATIENT(PatientID) DiseaseID → DISEASE(DiseaseID) (During treatment, a patient may be treated for different diseases)	DISEASE(<u>DiseaseID</u>, DName) PK: DiseaseID FK: -

Q5

E1(<u>A11</u>, A12, A13) PK: (A11, A12) All attributes: NOT NULL	E2(<u>A21</u>, A22, E1_A11, E1_A12) PK: A21 FK: (E1_A11, E1_A12) → E1(A11, A12) All attributes: NOT NULL
E3(<u>A31</u>, R4_A11, R4_A12, R5_A11, R5_A12, AR4) PK: A31 FK: (R4_A11, R4_A12) → E1(A11, A12) FK: (R5_A11, R5_A12) → E1(A11, A12) A31, AR4: NOT NULL R4_A11, R4_A12, R5_A11, R5_A12: NULL allowed	E4(<u>A41</u>, E2_A21, E3_A31) PK: A41 FK: E2_A21 → E2(A21) FK: E3_A31 → E3(A31) All attributes: NOT NULL
AR4(<u>AR4</u>) PK: AR4 All attributes: NOT NULL	

